

Chapter 4

Continuity and differentiation

4.1 Reference Books

There are many introductory calculus books that you could use for this part of the course. The Schaums Outline of Calculus book is an excellent one. It contains many worked examples and has lots of exercises for you to practice with and contains solutions to a large number of these.

- *Schaum's Outline of Calculus*, by Frank Ayres and Elliott Mendelson.

The Schaum's books are all reasonably priced and would be a good investment for you as it also contains lots of material that will be useful in subsequent years of your course. You can also access the full text of this book through MMU's library. Follow the links in the *Library* block on the unit's *Moodle* page to access the full text of the Schaum book and also to find the library catalogue entries for others.

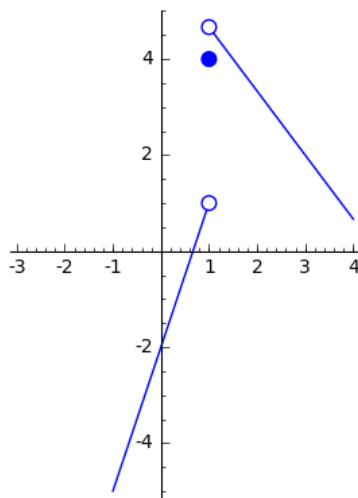
Two other textbooks that I recommend you consult are:

- *Modern Engineering Mathematics*, by Glyn James
- *Engineering Mathematics*, by K. A. Stroud

In these books you will find detailed explanations of the topics we will look at in the lectures as well as plenty of examples and exercises to practice with. It is a good idea to become familiar with both or either of these books as they (and their *Advanced* sequels) will be useful for the other mathematics units that you study.

4.2 Introduction

In this chapter we shall be studying functions whose co-domain is the set of real numbers $\mathbb{R}$. Usually the domain will be $\mathbb{R}$ as well although not always, as for instance with the function $y = \frac{1}{x}$ which has domain $\mathbb{R} \setminus \{0\}$. So we will not normally spell out the domain and co-domain of the functions we talk about. In any particular case the domain can be figured out by considering where the formula defining the function makes sense.

Figure 4.1: Discontinuous behaviour at of a function at $x = 1$.

4.3 Limiting values of functions

We have met the concept of a limit in the context of sequences and series. The concept is also useful when discussing real-valued functions. Let a and l be two real numbers. Informally we say that the function f tends to the limit l as x tends to a if the value $f(x)$ gets closer and closer to l as x gets closer and closer to a . We can represent this symbolically as

$$\lim_{x \rightarrow a} f(x) = l,$$

which we read as ‘the limit, as x tends to a , of $f(x)$ is equal to l ’. Alternatively we can write

$$f(x) \rightarrow l \text{ as } x \rightarrow a,$$

which we read as ‘ $f(x)$ tends to l as x tends to a ’.

Notice that this description makes no reference to the actual value (if any) of the function f at a . The concept of limit only refers to the behaviour of f as it *approaches* some limiting point a .

This definition is not sufficiently rigorous though. For instance, as the diagram in figure 4.1 illustrates, a function may have different limits depending on the direction from which x approaches a . If this situation arises then we can refer separately to the two different limits as follows. The limit of f as x tends to a from the left is written as

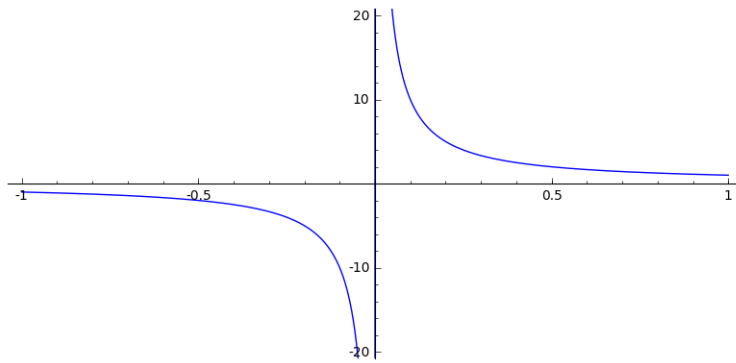
$$\lim_{x \rightarrow a^-} f(x),$$

and the limit of f as x tends to a from the right is written as

$$\lim_{x \rightarrow a^+} f(x).$$

A rigorous definition?

Just as in the case of the convergence for sequences, convergence and limiting values of functions also require a detailed ϵ -type definition. We shall postpone this for now in our discussion but we will meet the full detail when we discuss continuity later on.

Figure 4.2: The function $g(x) = 1/x$.

Although it is *not* an element of the real numbers $\mathbb{R}$ we can admit the concept of infinity (∞) into this definition as it will allow us to describe the behavior of functions in the following way. If a function f grows without bound as x approaches a then we can write

$$\lim_{x \rightarrow a} f(x) = \infty.$$

For example

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty,$$

and

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty,$$

If the values of a function f approach some limit l as the argument x gets larger and larger then we could write

$$\lim_{x \rightarrow \infty} f(x) = l.$$

For example

$$\lim_{x \rightarrow +\infty} \frac{1}{x} = 0,$$

and

$$\lim_{x \rightarrow -\infty} \frac{1}{x} = 0.$$

These limiting values of the function $y = \frac{1}{x}$ can be appreciated from its graph shown in figure 4.2. To calculate and work with limiting values of functions we can use the standard techniques that we introduced when dealing with limits of sequences and series. We have the following theorem

Theorem 4.1 (algebra of limits). *Let $l, m \in \mathbb{R}$ and suppose that $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$, then*

$$\begin{aligned}\lim_{x \rightarrow a} (\alpha f(x) + \beta g(x)) &= \alpha l + \beta m, \\ \lim_{x \rightarrow a} (f(x)g(x)) &= lm, \\ \lim_{x \rightarrow a} \frac{f(x)}{g(x)} &= \frac{l}{m}, \text{ providing } m \neq 0.\end{aligned}$$

Of particular use is l'Hopital's rule. This is used to evaluate the limit, $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$, of a quotient of functions f and g . Sometimes these limits can not be evaluated because it works out to some indeterminate form, i.e. $\frac{0}{0}, \frac{\infty}{\infty}$. In these cases we can differentiate both functions and then look at the limit of the resulting quotient. If this new limit exists then the rule says that it is equal to the original limit.

Theorem 4.2 (l'Hopital's rule). *If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$ or $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = \infty$ then providing the limit exists*

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)},$$

where f' and g' are the derivatives of f and g .

It may be necessary to apply l'Hopital's rule a number of times (i.e. taking second or third derivatives etc) in order to convert the original indeterminate limit into a determinate one.

Example 4.1. The following examples illustrate the use of these techniques.

1.

$$\lim_{x \rightarrow \infty} \frac{3x + 1}{2x} = \lim_{x \rightarrow \infty} \frac{3 + \frac{1}{x}}{2} = \frac{3}{2}.$$

2.

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1.$$

3.

$$\lim_{x \rightarrow \infty} \frac{x + 1}{x} = \lim_{x \rightarrow \infty} \frac{1 + \frac{1}{x}}{1} = 1.$$

4.

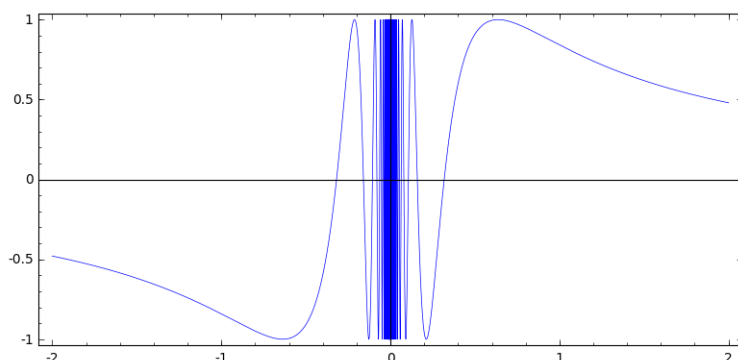
$$\lim_{x \rightarrow 0} \frac{\sin 2x}{3x} = \lim_{x \rightarrow 0} \frac{2 \cos 2x}{3} = \frac{2}{3}.$$

5.

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{x} = \lim_{x \rightarrow 0} \frac{e^x + e^{-x}}{1} = 2.$$

6.

$$\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3} = \lim_{x \rightarrow 0} \frac{\cos x - 1}{3x^2} = \lim_{x \rightarrow 0} \frac{-\sin x}{6x} = \lim_{x \rightarrow 0} \frac{-\cos x}{6} = \frac{-1}{6}.$$

Figure 4.3: The graph of $\sin(1/x)$ near $x = 0$.

4.4 Continuity

In the previous section we made the point that the limiting value, $\lim_{x \rightarrow a} f(x)$, of the function f does not depend on the actual value, $f(a)$, of f at the limiting point a . However it is an important feature (continuity) of a function when the limiting value actually equals the value at the limiting point. We give the following informal definition for continuity.

Definition 4.1 (Continuity). The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is *continuous* at the point $x = a$ if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

If f is continuous at *all* points of its domain then we say that f is a *continuous function*. A function that is not continuous is called *non-continuous*.

If this definition is not satisfied at a point $x = a$ then we say that the function f has a *discontinuity* at $x = a$.

Functions can be discontinuous in a number of different ways. For instance the left and right hand limits may differ at a point as we saw earlier. Another type of discontinuity is the rapidly oscillating behaviour near $x = 0$ of the function $y = \sin(\frac{1}{x})$ seen in figure 4.3.

Note that the plotting program struggles to represent the behavior accurately at $x = 0$ here. The point is that the limit $\lim_{x \rightarrow 0} \sin(\frac{1}{x})$ does not exist so even if the function were to be given a definition at $x = 0$ it could never be continuous there.

The continuity property is preserved by the usual operations on functions, as described in the following theorems.

Theorem 4.3 (Continuity of sums, products and quotients). *Suppose that functions f and g are both continuous at a point $x = a$, then for any $\alpha, \beta \in \mathbb{R}$ the linear combination function $\alpha f + \beta g$ is also continuous at $x = a$. Also the product function fg will be continuous at $x = a$ as well as the quotient function $\frac{f}{g}$ provided that g is non-zero in some surrounding interval of $x = a$.*

Theorem 4.4 (Continuity of composition). *Suppose that f and g are two functions whose domains and co-domains are such that the composition function*

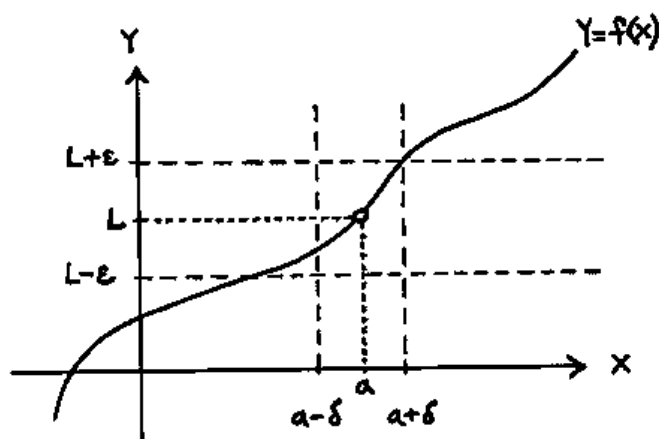
$g \circ f$ can be defined. If f is continuous at a point $x = a$ and g is continuous at the point $x = f(a)$ then the composition function $g \circ f$ is continuous at $x = a$.

The definition of continuity (and limiting values of functions) that we have given do lack rigor in that they rely on the notion of ‘*getting closer and closer to* ’ which we have not given a complete description of. A fully detailed rigorous treatment of continuity is the subject of *mathematical analysis* and is not possible within the scope of this unit. However we can give a taste of it by looking at the formal definition of continuity and how it is used to prove one of the cases of the above theorem.

Definition 4.2 (Formal definition of Continuity). The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is *continuous* at the point $x = a$ if and only if for all $\epsilon > 0$ there exists $\delta > 0$ such that if $|x - a| < \delta$ then $|f(x) - f(a)| < \epsilon$. In formal symbolic terms this becomes: f is continuous at a if and only if the following statement holds

$$\forall \epsilon > 0 \exists \delta > 0 |x - a| < \delta \Rightarrow |f(x) - f(a)| < \epsilon.$$

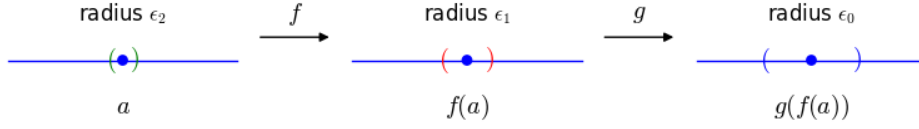
The following diagram is useful for picturing the role of the variables ϵ and δ in the definition of continuity.



Using this definition we can give rigorous proofs of the continuity theorems above. You will see a lot of similarity in the arguments here and the proofs of the similar properties of convergence of sequences. Those previous arguments are adapted to the new context and where δ -arguments take the place of the N -arguments.

Theorem 4.3 (Continuity of sums, products and quotients) Suppose that functions f and g are both continuous at a point $x = a$, then for any $\alpha, \beta \in \mathbb{R}$ the linear combination function $\alpha f + \beta g$ is also continuous at $x = a$. Also the product function fg will be continuous at $x = a$ as well as the quotient function $\frac{f}{g}$ provided that g is non-zero in some surrounding interval of $x = a$.

Proof. First we prove that continuity is preserved by linear combinations. Let $\epsilon > 0$ be given. There exists $\delta_1, \delta_2 > 0$ such that for all $|x - a| < \delta_1$ we have $|f(x) - f(a)| < \epsilon/(2|\alpha|)$, and for all $|x - a| < \delta_2$ we have $|g(x) - g(a)| < \epsilon/(2|\beta|)$.

Figure 4.4: The composition function $g \circ f$

Let $\delta = \min(\delta_1, \delta_2)$ and then for all $|x - a| < \delta$ we have

$$\begin{aligned} |(\alpha f + \beta g)(x) - (\alpha f + \beta g)(a)| &\leq |\alpha||f(x) - f(a)| + |\beta||g(x) - g(a)| \\ &< |\alpha|\epsilon/(2|\alpha|) + |\beta|\epsilon/(2|\beta|) \\ &= \epsilon. \end{aligned}$$

Therefore the definition is satisfied and the linear combination $\alpha f + \beta g$ is continuous at $x = a$ as required.

Next we prove that continuity is preserved by products. Let $\epsilon > 0$ be given. By continuity there exist $\delta_1, \delta_2 > 0$ such that for all $|x - a| < \delta_1$ we have $|f(x) - f(a)| < \min(\sqrt{\epsilon}/\sqrt{3}, \epsilon/(3|g(a)|))$ and $|x - a| < \delta_2$ we have $|g(x) - g(a)| < \min(\sqrt{\epsilon}/\sqrt{3}, \epsilon/(3|f(a)|))$.

Letting $\delta = \min(\delta_1, \delta_2)$ and then for all $|x - a| < \delta$ we have

$$\begin{aligned} |(fg)(x) - (fg)(a)| &\leq |f(x) - f(a)||g(x) - g(a)| + |g(a)||f(x) - f(a)| + |f(a)||g(x) - g(a)| \\ &< \sqrt{\frac{\epsilon}{3}}\sqrt{\frac{\epsilon}{3}} + |g(a)|\frac{\epsilon}{3|g(a)|} + |f(a)|\frac{\epsilon}{3|f(a)|} \\ &= \epsilon. \end{aligned}$$

Therefore the definition is satisfied and the product fg is continuous at $x = a$ as required.

The proof that quotients preserve continuity is again similar to the case for convergence of sequences. The details are left as an exercise for the reader. $\square$

Theorem 4.4 (Continuity of composition) *Suppose that f and g are two functions whose domains and co-domains are such that the composition function $g \circ f$ can be defined. If f is continuous at a point $x = a$ and g is continuous at the point $x = f(a)$ then the composition function $g \circ f$ is continuous at $x = a$.*

Proof. The conditions of the theorem are illustrated in figure 4.4. There we see illustrated three copies of the real number line with the action of the functions from left to right with f going first and then g applied to the result of f .

Let $\epsilon_0 > 0$ be given. As g is continuous at $f(a)$ there exists a radius ϵ_1 such that for all $|y - f(a)| < \epsilon_1$ we have $|g(y) - g(f(a))| < \epsilon_0$. But since f is continuous at a there exists a radius ϵ_2 such that for all $|x - a| < \epsilon_2$ we have $|f(x) - f(a)| < \epsilon_1$.

Combining these two gives us the required result, namely that for all $|x - a| < \epsilon_2$ we have $|g(f(x)) - g(f(a))| < \epsilon_0$, i.e. the composition $g \circ f$ is continuous at $x = a$. $\square$

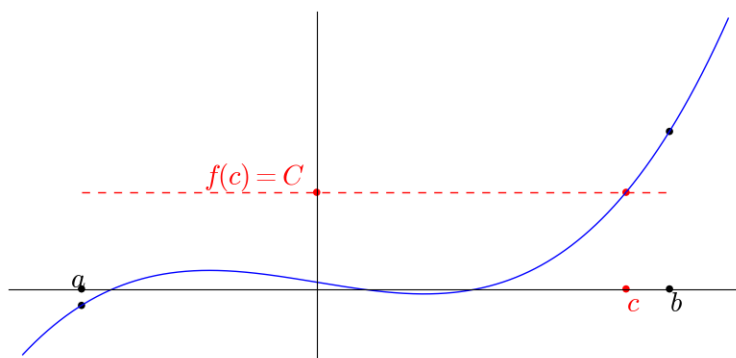


Figure 4.5: The Intermediate Value Theorem.

Importance of continuity

Mathematicians sometimes talk of a concept like continuity as describing the *well-behaved* nature of certain functions $\mathbb{R} \rightarrow \mathbb{R}$. So a well-behaved function is one that has nice reasonable properties, properties usually enjoyed by the real physical properties of the processes that we are using the functions to model for instance. A badly-behaved function $\mathbb{R} \rightarrow \mathbb{R}$ might be one that has no discernable reasonable properties or interpretation, such as some kind of random association of real numbers to other real numbers (though such things and their properties might still be of great mathematical interest!).

The concept of continuity generalizes to functions between lots of other kinds of mathematical space, such as say functions from $\mathbb{R}^n \rightarrow \mathbb{R}^m$, where n, m are positive integers. These are functions between different dimensional Euclidean geometric Euclidean spaces. Or perhaps functions between different Riemannian manifolds. A Riemannian manifold generalization of Euclidean space, i.e. a space which has a different kind of *curved* geometry rather than the *flat* geometry of Euclidean spaces. We can even study the concept of continuity for functions, and distance like measures, on spaces that have no kind of immediate intuitive geometric interpretation for us. Such topics are explored in the subject areas of *general topology* and *metric spaces*.

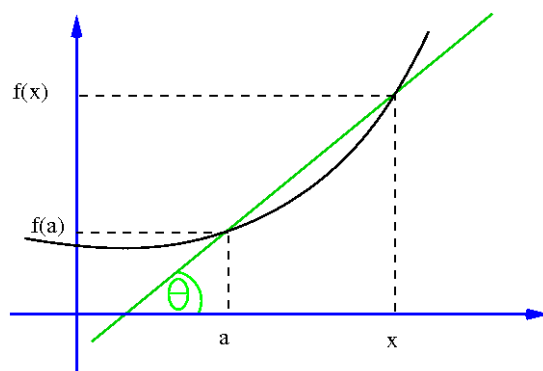
An example of continuity embodying the notion of *well-behaved* function we can consider the Intermediate Value Theorem which is illustrated in figure 4.5.

Theorem 4.5 (Intermediate Value Theorem). *Let f be a function defined on and continuous at all points in the closed interval $[a, b]$. Suppose that $f(a) \neq f(b)$. Then for any number C between $f(a)$ and $f(b)$, there exists a number $c \in [a, b]$ such that $f(c) = C$.*

The proof of this theorem makes use of the completeness or least upper-bound property of the real numbers (see appendix ??) and can be viewed as an embodiment of that property

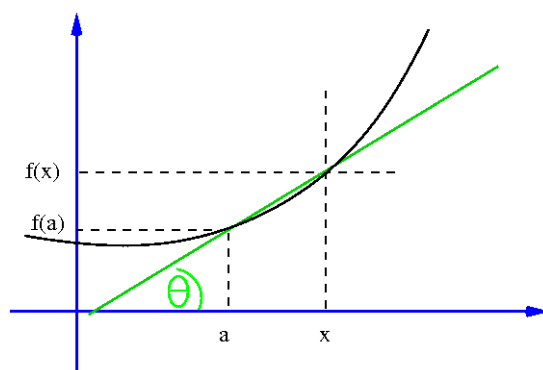
4.5 Differentiation

The derivative of a function is a quantity that measures the instantaneous rate of change of a function. It is defined using the notion of the limiting value of a certain quotient involving the function under consideration. This definition is motivated and explained by the following diagrams.

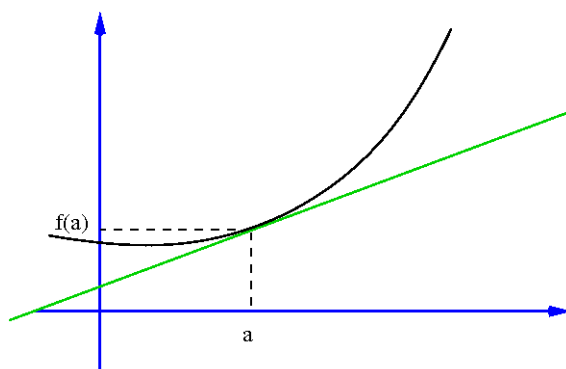


Change in value of a function f is measured by the gradient of the line joining two points on the graph of f .

$$\text{gradient} = \tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{f(x) - f(a)}{x - a}$$



As $x \rightarrow a$ the gradient describes the rate of change of f at the point $x = a$ more accurately.



In the limit as $x \rightarrow a$ the gradient line approaches the *tangent line* to the graph of f at the point $x = a$. The tangent line to a point on the graph is a line that, in a small neighborhood of the point, only intersects the graph at that point. The slope of the tangent line is the same as the slope of the graph at that point.

Definition 4.3 (Differentiable). A function f is *differentiable* at $x = a$ if f is continuous at $x = a$ and the following limit exists

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

If f is differentiable at $x = a$ then the value of the limit above is called the *derivative of f at $x = a$* and can be written symbolically as

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a},$$

or

$$\left. \frac{df}{dx} \right|_{x=a} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

Example 4.2. Consider the functions f and g defined by $f(x) = kx$ and $g(x) = kx^2$ where k is some constant. Find the derivatives of f and g at $x = a$ from first principles. ('*from first principles*' means working with the above definition of the derivative in terms of limits)

$$\begin{aligned} \left. \frac{df}{dx} \right|_{x=a} &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\ &= \lim_{x \rightarrow a} \frac{kx - ka}{x - a} \\ &= \lim_{x \rightarrow a} \frac{k(x - a)}{x - a} = k. \end{aligned}$$

The function g can be handled in a similar way.

$$\begin{aligned} \left. \frac{dg}{dx} \right|_{x=a} &= \lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a} \\ &= \lim_{x \rightarrow a} \frac{kx^2 - ka^2}{x - a} \\ &= \lim_{x \rightarrow a} \frac{k(x^2 - a^2)}{x - a} \\ &= \lim_{x \rightarrow a} \frac{k(x - a)(x + a)}{x - a} = 2ka. \end{aligned}$$

As is usually the case for us, a function f will be defined by a single formula in a variable x say and so the derivative of f at all points in its domain can be worked out in one go to give a formula in x which defines the *derivative function* of f . So given a function f we can speak of its derivative (function) $\frac{df}{dx}$ which when evaluated at a point $x = a$ give the *derivative of f at $x = a$* as defined by our earlier definition.

Higher derivatives

Once a function f has been differentiated to give the derivative $\frac{df}{dx}$ it can be differentiated again to give higher derivatives

$$\frac{d^2f}{dx^2} = \frac{d}{dx} \left(\frac{df}{dx} \right),$$

$$\frac{d^3f}{dx^3} = \frac{d}{dx} \left(\frac{d^2f}{dx^2} \right), \text{ etc....}$$

Standard derivatives

On the following two pages you will find tables showing the derivatives of some standard functions. These tables are taken from the Formula Handbook that you will have available in your summer exam.

1.4 Calculus

The constant of integration has been omitted.

$\int y \, dx$	y	$\frac{dy}{dx}$
$\left. \begin{array}{l} \frac{x^{n+1}}{n+1} \quad n \neq -1 \\ \log_e(x) \quad n = -1 \end{array} \right\}$	x^n	nx^{n-1}
$x \log_e(x) - x$	$\log_e(x)$	$\frac{1}{x}$
$\frac{1}{a} e^{ax}$	e^{ax}	ae^{ax}
$\frac{a^x}{\log_e(a)}$	$a^x \quad (a > 0)$	$a^x \log_e(a)$
$-\frac{1}{a} \cos(ax)$	$\sin(ax)$	$a \cos(ax)$
$\frac{1}{a} \sin(ax)$	$\cos(ax)$	$-a \sin(ax)$
$\frac{1}{a} \log_e[\sec(ax)]$	$\tan(ax)$	$a \sec^2(ax)$
$\log_e \left[\tan \left(\frac{x}{2} \right) \right]$	$\operatorname{cosec}(x)$	$-\operatorname{cosec}(x) \cot(x)$
$\log_e[\sec(x) + \tan(x)]$	$\sec(x)$	$\sec(x) \tan(x)$
$\log_e[\sin(x)]$	$\cot(x)$	$-\operatorname{cosec}^2(x)$
$\cosh(x)$	$\sinh(x)$	$\cosh(x)$
$\sinh(x)$	$\cosh(x)$	$\sinh(x)$
$\sin^{-1}(x)$	$\frac{1}{\sqrt{1-x^2}}$	
$\cos^{-1}(x)$	$\frac{1}{\sqrt{x^2-1}}$	
$\tan^{-1}(x)$	$\frac{1}{1+x^2}$	
$\sec^{-1}(x)$	$\frac{1}{x\sqrt{x^2-1}}$	
$\sinh^{-1}(x)$	$\frac{1}{\sqrt{x^2+1}}$	
$\log_e(x + \sqrt{x^2+1})$		
$\cosh^{-1}(x)$	$\frac{1}{\sqrt{x^2-1}}$	
$\log_e(x + \sqrt{x^2-1})$		
$\tanh^{-1}(x)$	$\frac{1}{1-x^2}$	

$\int y \, dx$	y
$\sin^{-1} \left(\frac{x}{a} \right)$	$\frac{1}{\sqrt{a^2 - x^2}}$
$\cos^{-1} \left(\frac{x}{a} \right)$	$-\frac{1}{\sqrt{a^2 - x^2}}$
$\tan^{-1} \left(\frac{x}{a} \right)$	$\frac{a}{a^2 + x^2}$
$\sec^{-1} \left(\frac{x}{a} \right)$	$-\frac{a}{x\sqrt{a^2 - x^2}}$
$\sinh^{-1} \left(\frac{x}{a} \right)$ or $\log_e \left(\frac{x + \sqrt{x^2 + a^2}}{a} \right)$	$\frac{1}{\sqrt{a^2 + x^2}}$
$\cosh^{-1} \left(\frac{x}{a} \right)$ or $\log_e \left(\frac{x + \sqrt{x^2 - a^2}}{a} \right)$	$-\frac{1}{\sqrt{a^2 - x^2}}$
$\tanh^{-1} \left(\frac{x}{a} \right)$	$\frac{a}{a^2 - x^2}$

Multiplication by a scalar

If $y = ku(x)$ where k is a scalar then

$$\frac{dy}{dx} = k \frac{du}{dx}$$

Product rule

If $y = u(x)v(x)$ then

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Quotient rule

If $y = \frac{u(x)}{v(x)}$ then

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Chain rule

If y is a function of u and u is the function of x then

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Sum/difference rules

$$\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$$

We can find derivatives of more complex functions which are built out of standard functions by using these known derivatives along with some properties of the derivative, which we now describe.

4.6 Properties of the derivative

Theorem 4.6 (Linearity). *Consider the linear combination $f = \alpha g + \beta h$ of functions g and h , where $\alpha, \beta \in \mathbb{R}$. The derivative of f satisfies*

$$\frac{df}{dx} = \alpha \frac{dg}{dx} + \beta \frac{dh}{dx}.$$

Proof. This is a straightforward application of properties of the limiting process for functions. Suppose that g and h are both differentiable at a point $x = a$.

$$\begin{aligned} \alpha g'(a) + \beta h'(a) &= \alpha \lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a} + \beta \lim_{x \rightarrow a} \frac{h(x) - h(a)}{x - a}, \text{ by defn.} \\ &= \lim_{x \rightarrow a} \left(\alpha \frac{g(x) - g(a)}{x - a} + \beta \frac{h(x) - h(a)}{x - a} \right), \text{ by algebra of limits} \\ &= \lim_{x \rightarrow a} \left(\frac{(\alpha g + \beta h)(x) - (\alpha g + \beta h)(a)}{x - a} \right), \text{ rearranging fractions} \\ &= (\alpha g + \beta h)'(a), \text{ by definition} \end{aligned}$$

The above equations demonstrate the linearity of the derivative. $\square$

Theorem 4.7 (Product rule). *Consider the function $f = gh$ which is a product of the two functions g and h . The derivative of f satisfies*

$$\frac{df}{dx} = g \frac{dh}{dx} + h \frac{dg}{dx}.$$

Proof. Suppose that g and h are both differentiable at a point $x = a$.

$$\begin{aligned} f'(a) &= \lim_{x \rightarrow a} \frac{g(x)h(x) - g(a)h(a)}{x - a}, \text{ def. of derivative and } f \\ &= \lim_{x \rightarrow a} \frac{g(x)(h(x) - h(a)) + h(a)(g(x) - g(a))}{x - a}, \text{ preparatory manipulation} \\ &= \left(\lim_{x \rightarrow a} g(x) \right) \left(\lim_{x \rightarrow a} \frac{h(x) - h(a)}{x - a} \right) + h(a) \left(\lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a} \right), \text{ alg. of limits} \\ &= g(a)h'(a) + h(a)g'(a), \text{ known derivs.} \end{aligned}$$

This is the required result. $\square$

Example 4.3. 1. Find the derivative of $y = (x^2 - 1) \sin x$. This is the product of the two functions

$$u(x) = x^2 - 1 \text{ and } v(x) = \sin x,$$

whose derivatives are

$$\frac{du}{dx} = 2x \text{ and } \frac{dv}{dx} = \cos x,$$

Which direction?

Strictly speaking we should read this chain of equations from the bottom up – for we shouldn't really speak of $f'(a)$ as being equal to anything before we've shown that it does indeed exist! Do you agree?

and so the required derivative is

$$\begin{aligned}\frac{dy}{dx} &= u \frac{dv}{dx} + v \frac{du}{dx} \\ &= (x^2 - 1) \cos x + 2x \sin x.\end{aligned}$$

2. Find the derivative of $y = xe^x$

Linear approximations and associated errors

Before moving on to consider other properties of differentiation we introduce another viewpoint on the derivative that will be needed for future proof arguments. Consider again the definition of the derivative of a function f at a point a in its domain.

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

We reformulate this slightly by writing x as $x = a + k$ and having the limiting process be as $k \rightarrow 0$, as follows

$$f'(a) = \lim_{k \rightarrow 0} \frac{f(a + k) - f(a)}{k}.$$

This suggests the existence of a function ϵ that measures the discrepancy between $f'(a)$ and the relative change quotient $(f(a + k) - f(a))/k$, defined as follows

$$\epsilon(k) = \frac{f(a + k) - f(a)}{k} - f'(a),$$

and this new function ϵ has the important convergence property

$$\lim_{k \rightarrow 0} \epsilon(k) = 0.$$

We can now rearrange the definition of ϵ to give the interesting expression

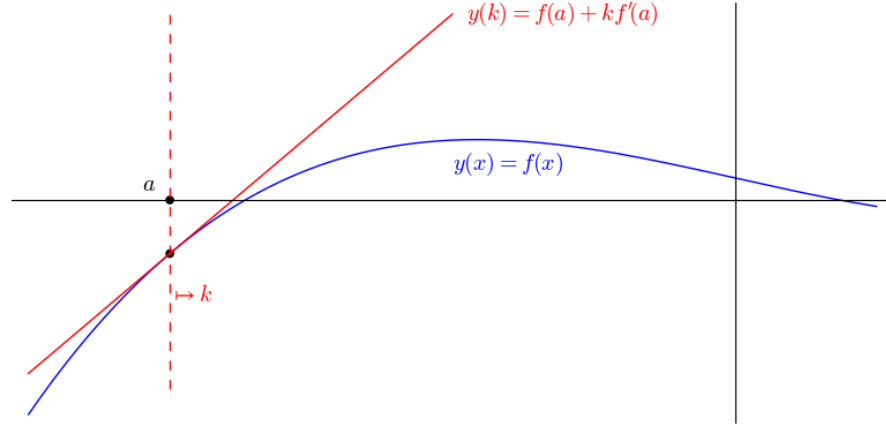
$$f(a + k) = f(a) + f'(a)k + k\epsilon(k).$$

This equation shows the *linear approximation*, $f(a) + kf'(a)$, to the value, $f(a + k)$ of the function at a point close to a , with the associated *error term* $k\epsilon(k)$. The linear approximation is the linear function (i.e. line) that best approximates the function in the region around the point $x = a$. Not only does the error-term approach zero as $k \rightarrow 0$ (as we would expect) but the error term divided by k , i.e. $\epsilon(k)$, converges to 0 as $k \rightarrow 0$.

Picturing these various quantities with the aid of figure 4.6 may help. It's important to appreciate how the linear approximation gives the equation of the tangent line to the graph of $f(x)$ at $x = a$ in the form of $y(k) = f(a) + kf'(a)$, where $k = x - a$.

This allows us to interpret the above to give an alternative definition of the derivative as follows:

Definition 4.4 (The derivative). The derivative, $f'(a)$, is the unique number that provides the linear approximation $f(a) + f'(a)k$ for the value $f(a + k)$ with the property that the error term $k\epsilon(k)$ has the convergence property that $\epsilon(k) \rightarrow 0$ as $k \rightarrow 0$.

Figure 4.6: The linear approximation to $f(x)$ at $x = a$.

This view of the derivative will allow us to give a rigorous proof of the chain rule. But it is also useful to have such slightly different views of the same concept. This notion of linear approximations (and other higher-order approximations) and the behaviour of their associated error terms is important in the subject of *numerical analysis*.

Theorem 4.8 (Chain rule). *Consider the composition $y = f \circ g$ of two functions f and g . Let $u = g(x)$ so that $y = f(u)$. Then the derivative of y satisfies*

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx},$$

or in the alternative notation, at a point $x = a$ we have

$$y'(a) = f'(g(a))g'(a).$$

Proof. We suppose that the function g is differentiable at the point a and that f is differentiable at the point $g(a)$. In the linear approximations view this means we have derivative quantities $g'(a)$ and $f'(g(a))$ and error functions ϵ and μ satisfying

$$\begin{aligned} g(a+k) &= g(a) + kg'(a) + k\epsilon(k) \\ f(g(a)+l) &= f(g(a)) + lf'(g(a)) + l\mu(l), \end{aligned}$$

with convergence properties $\epsilon(k) \rightarrow 0$ as $k \rightarrow 0$ and $\mu(l) \rightarrow 0$ as $l \rightarrow 0$.

Our task now is to prove the chain rule according to definition 4.4, i.e. to examine the quantity $y(a+k)$ and discover the linear approximation to it, showing that the derivative $y'(a)$ has the form described by the chain rule and that the error term has the necessary convergence property.

$$\begin{aligned} y(a+k) &= f(g(a+k)), \text{ defn. of composition} \\ &= f(g(a) + kg'(a) + k\epsilon(k)), \text{ from above for } g(a+k) \\ &= f(g(a)) + lf'(g(a)) + l\mu(l), \text{ from above with } l = kg'(a) + k\epsilon(k) \\ &= f(g(a)) + k(g'(a)f'(g(a))) + \lambda(k), \end{aligned}$$

where the error term $\lambda(k)$ is defined by

$$\lambda(k) = k\epsilon(k)f'(g(a)) + [kg'(a) + k\epsilon(k)]\mu(kg'(a) + k\epsilon(k)).$$

So we have established the required form for $y(a+k)$ which shows the expected derivative $y'(a)$ as asserted by the chain rule. To complete the proof we have just to prove that $\lambda(k)/k \rightarrow 0$ as $k \rightarrow 0$. This can be seen by considering the equation

$$\frac{\lambda(k)}{k} = \epsilon(k)f'(g(a)) + [g'(a) + \epsilon(k)]\mu(kg'(a) + k\epsilon(k)).$$

We can prove that $\lambda(k)/k \rightarrow 0$ by applying the algebra of limits theorem to the above, noting that $kg'(a) + k\epsilon(k) \rightarrow 0$ as $k \rightarrow 0$.

Check the details
 □ Do take the time to confirm for yourself that $\lambda(k)/k$ has the required convergence property.

Example 4.4. The following examples illustrate the use of the chain rule.

1. Differentiate $y = (x^2 - 1)^5$.

Let $u = x^2 - 1$ so that $y = u^5$. Then $\frac{du}{dx} = 2x$ and $\frac{dy}{du} = 5u^4$ and so

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = 5u^4 2x = 5(x^2 - 1)^4 2x = 10x(x^2 - 1)^4.$$

2. Differentiate $y = \sin(2x^2 - x)$.

Let $u = 2x^2 - x$ so that $y = \sin u$. Then $\frac{du}{dx} = 4x - 1$ and $\frac{dy}{du} = \cos u$ and so

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = \cos(u)(4x - 1) = (4x - 1) \cos(2x^2 - x).$$

To complete the collection of basic properties of the derivative we need to know how to differentiate functions that are defined as quotients of other functions, i.e. how the derivative of the quotient depends on the derivatives of the numerator and denominator of the quotient. This is provided by the well-known quotient rule.

Theorem 4.9 (Quotient rule). *If y is the quotient of two functions $y = \frac{u}{v}$ then*

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}.$$

Or in the alternative notation

$$y'(a) = \frac{v(a)u'(a) - u(a)v'(a)}{v(a)^2}.$$

The details of the proof of the quotient rule are left as an exercise. One way to prove it is to make use of the chain rule and product rule properties that we have already established together with the following lemma.

Lemma 4.10. *Consider the reciprocal function r defined on the non-zero reals, $\mathbb{R} \setminus \{0\}$, by $r(x) = 1/x$. The derivative of r at a point $a \neq 0$ is given by*

$$r'(a) = \frac{-1}{a^2}.$$

Proof. Exercise. □

Example 4.5. These examples illustrate the use of the quotient rule.

1. Differentiate $y = \frac{x^2-1}{\sin x}$.

Let $u = x^2 - 1$ and $v = \sin x$ then $\frac{du}{dx} = 2x$ and $\frac{dv}{dx} = \cos x$. Then

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} = \frac{2x \sin x - (x^2 - 1) \cos x}{\sin^2 x}.$$

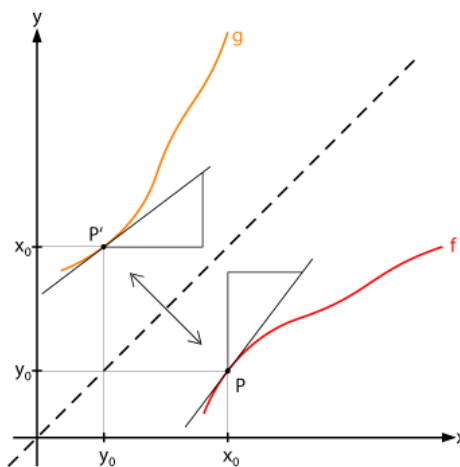
2. Differentiate $y = \frac{2x-1}{3x^2+x}$.

Let $u = 2x - 1$ and $v = 3x^2 + x$ then $\frac{du}{dx} = 2$ and $\frac{dv}{dx} = 6x + 1$. Then

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} = \frac{2(3x^2 + x) - (2x - 1)(6x + 1)}{(3x^2 + x)^2} = \frac{-6x^2 + 6x + 1}{(3x^2 + x)^2}.$$

Derivatives of inverse functions

There is a nice relationship between the derivative of a function and its inverse which we can appreciate from the diagram below. Shown below is the graph of the function $y = f(x)$ and also the graph of the inverse function $x = g(y)$. These two graphs are related by a reflection in the line given by $y = x$.



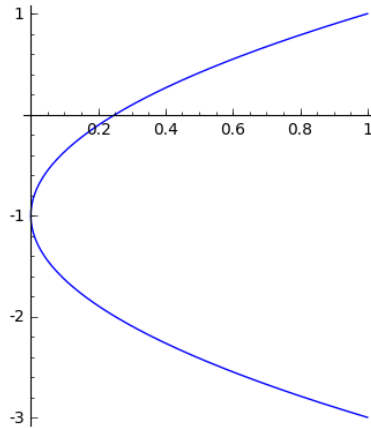
This leads to the following relationship for the derivative of a function and its inverse

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}.$$

Parametric differentiation

Curves that are not graphs of functions can often be specified by a pair of equations

$$x = x(t), y = y(t),$$

Figure 4.7: The parametric curve $(t^2, 2t - 1)$.

which describe how the coordinates (x, y) of a point on the curve relate to a variable t . These equations are called the *parametric* equations for the curve and the variable t is called the *parameter*. As the value of t changes the point $(x(t), y(t))$ traces out the curve.

Example 4.6. These examples illustrate the technique of parametric differentiation.

1. The parametric curve $x = t^2$, $y = 2t - 1$ with $t \in [-1, 1]$ is shown in figure 4.7.
2. The parametric curve

$$x = \cos t - \cos(2t), \quad y = \sin t - \sin(2t), \quad t \in [-\pi, \pi]$$

is shown in figure 4.8.

We can still talk of the tangent lines to these curves and using the chain rule we can work out the derivative $\frac{dy}{dx}$ for a parametric curve. We think of y as being a function of x , $y = y(x)$ as well as being a function of t , $y = y(t)$. Applying the chain rule in the former case gives

$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt},$$

and so

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{1}{\frac{dx}{dt}},$$

which using the inverse function rule we can also write as

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx}.$$

By carefully applying the chain and inverse function rule again we can show that

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{1}{\frac{dx}{dt}}.$$

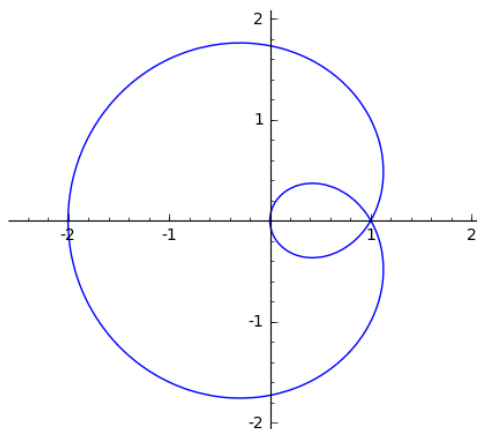


Figure 4.8: The parametric curve $(\cos(t) - \cos(2t), \sin(t) - \sin(2t))$.

Example 4.7. More parametric examples.

1. For the last example above we have

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{1}{\frac{dx}{dt}} = \frac{\cos t - 2 \cos(2t)}{-\sin t + 2 \sin(2t)}.$$

2. If a parametric curve is described by $x = te^t$ and $y = e^t$ determine $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in terms of t .

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{1}{\frac{dx}{dt}} = \frac{e^t}{te^t + e^t} = \frac{1}{t+1},$$

and

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{1}{\frac{dx}{dt}} = \frac{-1}{(t+1)^2} \frac{1}{te^t + e^t} = \frac{-1}{e^t(t+1)^3}.$$

Implicit differentiation

Suppose that a variable y is a function of x , i.e. $y = y(x)$, but that we are not given this functional relationship *directly* in the form of a formula for y in terms of x , but rather the function is described *indirectly* or *implicitly* by an equation involving y and x , such as

$$x^2 + y^2 = 1, \text{ or } x^2 - y^2 = 1, \text{ or } \sin(xy) = 1.$$

We can use the rules (chain, product, etc) developed so far to find an expression for the derivative $\frac{dy}{dx}$ in such cases. When the equation describing the relationship between y and x is non-trivial we would expect the resulting expression for $\frac{dy}{dx}$ to depend on the value of the function y as well as on the variable x .

Example 4.8. Find $\frac{dy}{dx}$ for the function $y = y(x)$ defined implicitly by the equation

$$x^2 + 2y^2 - xy + 2x = 4.$$

We proceed by differentiating both sides of the equation with respect to x and carefully using the chain rule and product rule.

$$\begin{aligned} x^2 + 2y^2 - xy + 2x &= 4 \\ \Rightarrow \frac{d}{dx}(x^2 + 2y^2 - xy + 2x) &= \frac{d}{dx}(4) \\ \Rightarrow 2x + 4y \frac{dy}{dx} - x \frac{dy}{dx} - y + 2 &= 0 \\ \Rightarrow \frac{dy}{dx}(4y - x) &= y - 2 - 2x \\ \Rightarrow \frac{dy}{dx} &= \frac{y - 2 - 2x}{4y - x}. \end{aligned}$$

Logarithmic differentiation

This refers to the technique we use to differentiate a function $y = y(x)$ say where the variable x appears in the power position in the formula for y .

Example 4.9. Differentiate $y = x^x$.

To begin with we take the logarithm of both sides of the defining equation and then use the technique of implicit differentiation.

$$\begin{aligned} y &= x^x \\ \Rightarrow \ln y &= \ln(x^x) \\ \Rightarrow \ln y &= x \ln x \\ \Rightarrow \frac{d}{dx}(\ln y) &= \frac{d}{dx}(x \ln x) \\ \Rightarrow \frac{1}{y} \frac{dy}{dx} &= x \frac{1}{x} + \ln x = 1 + \ln x \\ \Rightarrow \frac{dy}{dx} &= y(1 + \ln x) = x^x(1 + \ln x) \end{aligned}$$

Applications of differentiation

We shall consider four main applications of the technique of differentiation. They are

- finding and classifying stationary points of functions,
- rate of change problems,
- curve sketching,
- kinematics

But first we might briefly revisit the Taylor / Maclaurin series that we met in term 1 now that we have a fuller picture of differentiation and integration.

4.7 Taylor series and Maclaurin series

We saw earlier that under certain conditions Taylor's theorem provides for the existence of a *Taylor series* expansion of a function f . The Taylor series is the power series expansion of $f(x)$ about the point $x = a$ of the form

$$f(a+x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} x^n.$$

The Taylor series expansion of a function f about the point 0 is call the *Maclaurin series* expansion,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n.$$

Maclaurin series of some standard functions

$$(1+x)^r = 1 + rx + \frac{r(r-1)}{2!}x^2 + \frac{r(r-1)(r-2)}{3!}x^3 + \dots + \frac{r(r-1)\dots(r-n+1)}{n!}x^n + \dots$$

(where $-1 < x < 1$, $r \in \mathbb{R}$)

$$e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \dots + \frac{1}{n!}x^n + \dots, \quad (x \in \mathbb{R})$$

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \dots + \frac{(-1)^n}{(2n+1)!}x^{2n+1} + \dots, \quad (x \in \mathbb{R})$$

$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \dots + \frac{(-1)^n}{(2n)!}x^{2n} + \dots, \quad (x \in \mathbb{R})$$

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \dots + \frac{(-1)^n}{n+1}x^{n+1} + \dots, \quad (-1 < x \leq 1)$$

$$\tan x = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \frac{17}{315}x^7 + \dots, \quad \left(-\frac{\pi}{2} < x < \frac{\pi}{2}\right)$$

Exercises 4.1. Here are some exercises that explore Maclaurin series further.

1. Find the Maclaurin series expansion of $f(x) = e^x \sin x$ in terms of derivatives above.
2. Confirm your answer to 1) using the Maclaurin series for e^x and $\sin x$.
3. Obtain the expansion of $(1-x^2)^{-1/2}$ and deduce a power series expansion for $\sin^{-1}(x)$.

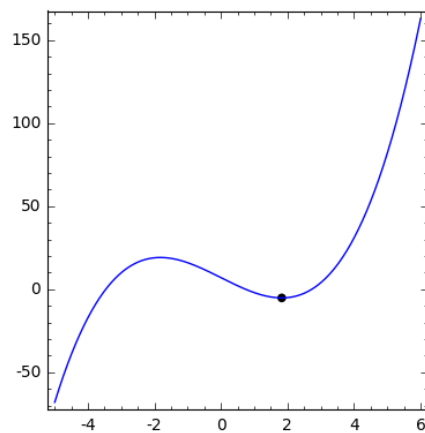


Figure 4.9: A local minimum

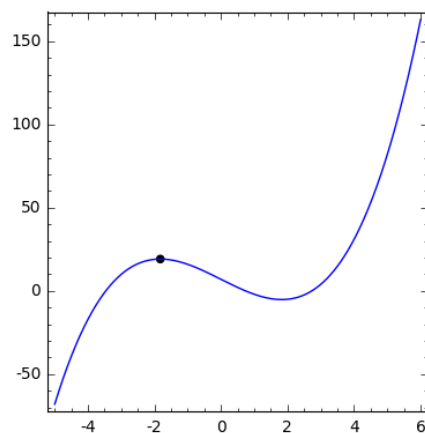


Figure 4.10: A local maximum

4.8 Stationary points of functions

Definition 4.5 (Stationary point). A *stationary point* (or *critical point*) of a function $f(x)$ is point on the graph $y = f(x)$ where the tangent to the graph has slope zero. That is, a stationary point is a solution a to the equation

$$\frac{dy}{dx} = 0.$$

Stationary points of functions of one variable can be of three types: a local maximum, local minimum or inflection point. See the diagrams in figures 4.9, 4.10 and 4.11 for illustrations of these different types.

Maximums and minimums are also sometimes called *turning points* of the function.

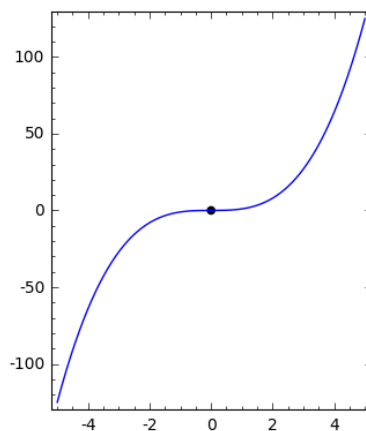


Figure 4.11: A point of inflection.

Classification of stationary points using derivatives

As we said a stationary point $x = a$ of the function $y = f(x)$ is a point satisfying

$$\left. \frac{dy}{dx} \right|_{x=a} = 0.$$

The point $x = a$ is a

- a **maximum** if

$$\left. \frac{dy}{dx} \right|_{x < a} > 0 \text{ and } \left. \frac{dy}{dx} \right|_{x > a} < 0,$$

or equivalently ,

$$\left. \frac{d^2y}{dx^2} \right|_{x=a} < 0.$$

- a **minimum** if

$$\left. \frac{dy}{dx} \right|_{x < a} < 0 \text{ and } \left. \frac{dy}{dx} \right|_{x > a} > 0,$$

or equivalently ,

$$\left. \frac{d^2y}{dx^2} \right|_{x=a} > 0.$$

- a **point of inflection** if either

$$\left. \frac{dy}{dx} \right|_{x < a} < 0 \text{ and } \left. \frac{dy}{dx} \right|_{x > a} < 0,$$

or ,

$$\left. \frac{dy}{dx} \right|_{x < a} > 0 \text{ and } \left. \frac{dy}{dx} \right|_{x > a} > 0.$$

Example 4.10. These examples illustrate the finding and classifying of stationary points.

1. Determine the turning points of the function $y = x^3 - x$.

The turning points are found by solving the equation

$$\frac{dy}{dx} = 3x^2 - 1 = 0.$$

This has solutions

$$x_1 = +\sqrt{\frac{1}{3}} \text{ and } x_2 = -\sqrt{\frac{1}{3}}.$$

Using the second derivative test to determine the type of each turning point we see that

$$\frac{d^2y}{dx^2} = 6x,$$

and so

$$\left. \frac{d^2y}{dx^2} \right|_{x=x_1} = 6x_1 \approx 3.462 > 0$$

and

$$\left. \frac{d^2y}{dx^2} \right|_{x=x_2} = 6x_2 \approx -3.462 < 0.$$

So we conclude that x_1 is a local minimum turning point and x_2 a local maximum turning point of the function $y = x^3 - x$.

2. Classify the stationary points of the function $y = 3x^4 - 4x^3$.

The equation

$$\frac{dy}{dx} = 12x^2(x - 1) = 0$$

has solutions $x_1 = 0$ and $x_2 = 1$. Now for points close to 0 we have

$$\left. \frac{dy}{dx} \right|_{x < 0} < 0 \text{ and } \left. \frac{dy}{dx} \right|_{x > 0} < 0$$

and therefore x_1 is a point of inflection. Check the second derivative $\left. \frac{d^2y}{dx^2} \right|_{x=x_2}$ to see that x_2 is a local minimum of the original function.

4.9 Curve Sketching

Recall that a *rational function* is a function f that is defined as a quotient of polynomials in x . For example the functions f defined by

$$f(x) = \frac{3x^2 - 7}{2x + 1}.$$

We considered these types of functions last term when we looked at their expansions by partial fractions. This technique was important for integrating such functions. What we want to be able to do now is to compose rough sketches of these functions which show the important features of their graphs. Admittedly these days we often turn to computers to produce accurate graphs. Nevertheless being able to do this ourselves with pen and paper gives us a

good understanding of the behavior of such functions and how the techniques of differentiation etc are used to produce these sketches.

To compose a sketch of such a function we consider various properties. The important ones are

- The domain of the function, i.e. are there any values of x for which $f(x)$ is undefined?
- The behaviour of $f(x)$ near these undefined points. Typically we will observe that $f(x)$ tends to $\pm\infty$ as x tends to one of these undefined points.
- The locations where the graph of f cuts the coordinate axes, i.e. the value $f(0)$ and the solution x to $f(x) = 0$.
- *Asymptotes* of f . These are lines (or other curves) that $f(x)$ approaches as x tends to $\pm\infty$.
- Stationary points of f . We identify any local/global minimum or maximum points and any points of inflections.
- Whether $f(x)$ is positive or negative on various parts of the domain. Whether $f(x)$ lies above or below any asymptotic lines/curves at various parts of the domain.

Example 4.11. Let us apply these considerations to the function f defined above by

$$f(x) = \frac{3x^2 - 7}{2x + 1}.$$

- f will be undefined when the denominator is zero, i.e. when $x = -\frac{1}{2}$. At all other points $f(x)$ is defined. So the domain of this function is $\mathbb{R} \setminus \{-\frac{1}{2}\}$.
- If x is close to $-\frac{1}{2}$ and less than $-\frac{1}{2}$ then the denominator is negative and the numerator is negative so $f(x)$ will be positive and as x tends to $-\frac{1}{2}$ then $f(x)$ tends to $+\infty$. On the other hand if x is close to $-\frac{1}{2}$ and greater than $-\frac{1}{2}$ then the denominator is positive and the numerator is negative so $f(x)$ will be negative and as x tends to $-\frac{1}{2}$ then $f(x)$ tends to $-\infty$.
- $f(0) = -7$ and $f(x) = 0$ when $x = \pm\sqrt{\frac{7}{3}} \approx \pm 1.53$.
- Consider the definition of f ,

$$f(x) = \frac{3x^2 - 7}{2x + 1}.$$

Dividing the numerator by the denominator as much as possible leads to the expression

$$f(x) = \frac{3}{2}x - \frac{3}{4} - \frac{\frac{25}{4}}{2x + 1}.$$

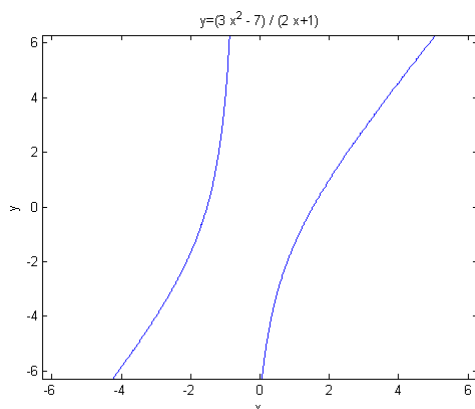
So we see that the line $y = \frac{3}{2}x - \frac{3}{4}$ is an asymptote of f . As x tends to $\pm\infty$ the term $\frac{\frac{25}{4}}{2x+1}$ tends to zero and so the points $y = f(x)$ move closer and closer to this line. As x tends to $+\infty$, $f(x)$ will tend to the line from below and as x tends to $-\infty$, $f(x)$ will tend to the line from above.

- To find the stationary points of f we solve the equation $\frac{df}{dx} = 0$. The derivative is (using the quotient rule)

$$\frac{df}{dx} = \frac{(2x+1)(6x) - (3x^2-7)(2)}{(2x+1)^2} = \frac{6x^2+6x+14}{(2x+1)^2}.$$

So we need to solve $6x^2 + 6x + 14 = 0$. But the discriminant $b^2 - 4ac$ of this equation is negative and so this equation has no real roots. Therefore f has no turning points.

Finally we can put all this information together to produce a sketch of the function



Exercises 4.2. Here are some more examples to investigate.

1.

$$y = \frac{x^2 + 1}{x^2 - 3x + 2}.$$

2.

$$y = \frac{x}{1 + x^2}.$$

3.

$$y = \frac{x^3}{2x - 1}.$$

4. *This last curve is not the graph of a rational function. But it illustrates some more sketching techniques.* Consider the curve defined by the equation

$$x^2 + xy + y^2 = 1.$$

Show that the maximum and minimum values taken by y are $\frac{\pm 2}{\sqrt{3}}$. By changing to polar coordinates sketch the curve. Polar coordinates are the pair of coordinates (r, θ) which describe the location of a point in the plane by specifying the points *radius* r , i.e. the distance from the point to the origin, and the angle θ that the radius line makes with the positive direction of the x -axis. The equations relating the Cartesian coordinates (x, y) and the polar coordinates (r, θ) are

$$x = r \cos \theta, \quad y = r \sin \theta.$$

4.10 Kinematics

Kinematics is the study of the *motion* of a body without reference to the forces acting on the body. We study the motion of a body in 1-dimension using the three functions: x displacement, v velocity and a acceleration.

- $x(t)$ is the position of the body at time t ,
- $v(t)$ is the velocity of the body at time t ,
- $a(t)$ is the acceleration of the body at time t .

These quantities are related by

$$v(t) = \frac{dx}{dt} \text{ and } a(t) = \frac{dv}{dt}.$$

Various equations relating the $x(t)$, $v(t)$, $a(t)$ and t can be found by integrating the above relations.

The simplest types of problem to study are *constant acceleration problems*. Here the acceleration is constant, e.g. due to the influence of gravity (9.8ms^{-2} downwards), or a car accelerating at a constant rate, etc.

Example 4.12. These simple kinematics problems can be solved by applying the relationships between position, velocity and acceleration.

1. A car waits at a traffic light. When the light turns green the car accelerates at a constant rate of 6ms^{-2} for 4 seconds. What velocity will the car have reached after 4 seconds and how far will it have traveled.
2. A student, who is running to catch his bus, sees the bus start to pull away from the stop. He is 20 m away from the stop and running at his top speed of 5ms^{-1} . If the bus accelerates at 2ms^{-2} , can he catch the bus?
3. Suppose you are at a window 18m above the ground and you want to throw a stone out the window so that it will be in the air for exactly 3 seconds. How should you throw it?

Exercises 4.3. These exercises involve the definitions, properties and applications of differentiation.

1. Decide if the following limits exist and if so evaluate them.

a)

$$\lim_{x \rightarrow 3} \frac{2x - 1}{3x + 2}$$

b)

$$\lim_{x \rightarrow 2} \frac{3x - 6}{2x}$$

c)

$$\lim_{x \rightarrow 0} \frac{2 \sinh x - 2 \sin x}{\cosh x - 1}$$

d)

$$\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{1 - \cos(2x)}$$

e)

$$\lim_{x \rightarrow 0} \frac{\sin(4x)}{\sin(6x)}$$

f)

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$$

g)

$$\lim_{x \rightarrow \infty} \frac{3x^3 - 6x}{3x + 2x^3}$$

h)

$$\lim_{x \rightarrow 0} \frac{5x - 6}{2x^3}$$

i)

$$\lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}}$$

2. Using the formal definition of continuity prove that if two functions f and g are both continuous at the point $x = a$ then so is the function $f + g$.
3. Prove from first principles that $\frac{d}{dx}(x^3) = 3x^2$.
4. Find $\frac{dy}{dx}$ for each of the following functions

a) $y = 5 \cos(2x) + 3e^{2x} - 1$

b) $y = e^{4x} \cos(2x)$

c) $y = \frac{\sin(2x)}{xe^x}$

d) $y = e^{\cos(3x)}$

e) $y = (2x)^{3x} + \sin(2x^2 - 7e^x)$

f) $y = \cos^{-1} x$

5. Find the gradients of the following implicitly defined functions y at the points indicated

a) $3x^2 + y^2 + 2xy - 6 = 0$ at the point $(1, 1)$

b) $x^2 + y^2 - 6xy + 4x + 4 = 0$ at the point $(1, 3)$

6. Given that

a) $x = 5 \cosh t$ and $y = 4 \sinh t$

b) $x = 2 \cos t$ and $y = \sin^2 t$

c) $x = t^2 - 4$ and $y = t^3 - 4t$ for $-3 \leq t \leq 3$

determine $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in terms of t for each parametric curve. Sketch these curves using MATLAB.

7. Find $\frac{dy}{dx}$ where $y = (x + 3)^{x^2 - 3}$.

8. Find $\frac{d}{dx}\left(\frac{uvw}{z}\right)$ where u, v, w and z are functions of x .

9. Suppose that an athletics field consists of a rectangle with semi-circular ends as shown. Determine the maximum possible area of the rectangular part of the field if the running track around the perimeter of the whole field is to have a length of exactly 400m.



10. Determine the dimensions of a cylindrical food tin of volume 330ml that has minimum surface area.
11. Sketch graphs of the following functions,

- a)

$$y = \frac{2 + x}{1 + x},$$

- b)

$$y = \frac{1}{2} \left(x + \frac{2}{x} \right),$$

- c)

$$y = \frac{3x^4 + 12x^2 - 4}{8x^3},$$

- d)

$$y = \frac{(x - 1)(x - 2)}{(x + 1)(x - 3)}.$$

12. By locating their turning points and asymptotes sketch graphs of the following functions,

a)

$$y = \frac{x^2 - 8x + 15}{x},$$

b)

$$y = \frac{x + 1}{x - 1},$$

c)

$$y = \frac{x^2 + 5x - 14}{x + 5}.$$

13. Consider the function f defined by

$$f(x) = e^{\cos x}.$$

Show that $f'(x) = -f(x) \sin x$ and hence obtain a formula for $f''(x)$ in terms of $f(x)$ and $f'(x)$. Differentiate again to find a formula for $f^{(3)}(x)$. Use these to find the first few terms of the Maclaurin series for f .

Confirm that your answer is correct by obtaining the Maclaurin series for f using the series for e^x and $\cos x$.

14. Show that the binomial expansion of $(1 + x)^{-1}$ is

$$(1 + x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \dots, \quad (-1 < x < 1).$$

Use this to obtain the Maclaurin series for $\tan^{-1} x$ which has can be defined by

$$\tan^{-1} x = \int_0^x \frac{1}{1 + z^2} dz.$$

15. Use the Maclaurin series for $\sin x$ and $\cos x$ to obtain the series for $\tan x$.
 16. Deduce the series for $\ln(\cos x)$ using the series for $\ln x$ and $\cos x$.
 17. Use the series for e^x and $\cos x$ to show that

$$\lim_{x \rightarrow 0} \left(\frac{e^x + e^{-x} - 2}{2 \cos 2x - 2} \right) = \frac{-1}{4}.$$

18. Use the series for e^{-x^2} to show that

$$\int_0^{1/2} e^{-x^2} dx \approx 0.461.$$

For these kinematics problems assume that the effects of air resistance can be ignored and that acceleration due to gravity is 9.8 ms^{-2} .

19. A stone is dropped from a height of 5m above the ground. How long will it take to drop to the ground? What is its speed upon impact?

20. A car starts from rest and accelerates in a straight line with constant acceleration. If it reaches a speed of 20ms^{-1} after 5s what is its acceleration and how far will it have travelled?
21. A projectile is fired straight up into the air. If the projectile takes 2.4s to reach its highest point what was its initial speed and how high did it reach?

Finding optimal solutions to problems by finding appropriate stationary points of functions is an important application of the technique of differentiation. Here are some further exercises for you to try.

22. Of all the rectangles having a particular area, which one has the smallest perimeter?
23. Of all the rectangles having a particular perimeter, which one has the greatest area?
24. A box with a square base and no top is to have a volume of 3m^3 . Find the dimensions of the box which uses the least amount of material in its construction.
25. Suppose that a rectangle has two corners lying on the x -axis and the other two lying on the parabola $y = 8 - x^2$. Find such a rectangle with maximal area.
26. What is the largest rectangle one can inscribe in a circle? That is, find the rectangle of largest area that can fit inside the circle of radius r .
27. Consider a river that is 0.5 miles wide and runs in a straight line for the purposes of this problem. A pipeline is to be laid across this river and run to a point 3 miles downstream on the opposite bank. Laying pipe under the water is expensive, in fact 25% more expensive per mile than laying pipe on dry land. How should the pipeline be laid across the river in order to minimize the total cost.